

## How a claim is scored: credence and faithfulness

**Credence:** will this claim survive an independent confirmatory test?    **Faithfulness:** does the model capture the mechanism, not just the statistics?    Green letters link each factor to an axis.

(a) Credence ladder (scoring v2, one per hypothesis)

1 • **What is scored** one clause: the card’s “**claim that stands**”, with its scope (periods, units, effect size). Withdrawn, post hoc or unpowered sub-results are named exclusions. A negative claim states the effect size at which its power reaches 0.8 and is scoped to it; below that it is “inconclusive”.

## 2 · Base rate $p_0$ from provenance

|                                                                               |                                                       |                                              |
|-------------------------------------------------------------------------------|-------------------------------------------------------|----------------------------------------------|
| <b>0.40</b> pre-registered primary prediction and estimator fixed before data | <b>0.30</b> secondary estimator redesigned after data | <b>0.20</b> post hoc claim found in the data |
|-------------------------------------------------------------------------------|-------------------------------------------------------|----------------------------------------------|

**3 · Multiply the odds  $p/(1 - p)$  by each factor that applies** (one design is credited once)

|          |                                                                    |               |                         |
|----------|--------------------------------------------------------------------|---------------|-------------------------|
| <b>C</b> | calibrated null passed (DQ8 sizes or own synthetic)                | ×2            | (mis-sized ×0.7)        |
| <b>I</b> | replication: same sign in $\geq 2/3$ of $\geq 6$ independent units | ×2            | (one unit ×0.6)         |
| <b>B</b> | robust to data versions, embedding models, estimators              | ×2            | (changed by a fix ×0.3) |
| <b>F</b> | identified: synthetic recovery at the real counts                  | ×1.5          | (not identifiable ×0.5) |
| <b>E</b> | consistent with a natural experiment (intervention)                | ×2            |                         |
| <b>H</b> | rival model beaten on held-out data                                | ×1.5          |                         |
| <b>G</b> | ground truth agrees (DQ6 roles, rooms, scheduler)                  | ×1.5          |                         |
| <b>I</b> | <b>locked holdout</b> passed / failed                              | ×5 / ×0.15    |                         |
|          | penalties: unvalidated LLM labels, problem inputs, multiplicity    | ×0.5–0.8 each |                         |

4 · **Credence**  $p = \text{odds}/(1 + \text{odds})$ , capped at 0.95 before a holdout run (0.98 after)

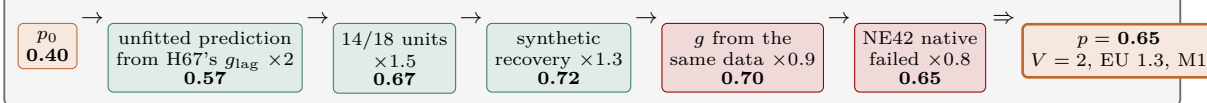
## 5 · Two raters

coordinator + a **blind** rater (never sees the other score)  $\rightarrow$  log-odds mean.  
 $|\Delta p| > 0.25$  or  $|\Delta V| > 1 \rightarrow$  adjudication  
 under rules 1–9 (score one clause; scope negatives by power; same-data agreement is robustness, not replication; ...)

## 6 · Use and depth

- value if true  $V$  (0-5): the decision it changes, magnitude in that decision's units, generality
- EU =  $pV$  (dashboard sorts by this)
- M0 pattern · M1 signature + rival rejected
- M2 intervention + rival + synthetic
- fragile**: headline moved across versions

Worked example: H111 (talk obeys the Fano sum rule in regime III)



Rules: writeup/hypothesis-pages/RUBRIC.md (scoring v2, adjudication rules 1-9) · STANDARDS.md (impostors, two layers) · paper.tex §Assessing model faithfulness. State as of 2026-10-04.

(b) Faithfulness scorecard (per model, mapping and window)

**Fit is not faithfulness.** A maximum-entropy model reproduces its fitted correlations by construction. Each axis scores  0 not done or failed,  1 partial,  2 passed, with evidence linked from the card.

**A** mapping

every variable defined from dataset fields; assumptions listed; invariant across model families and regimes

## assumptions

stationarity (split halves), Markov order, Hawkes time-rescaling, update-order audit; robust across data versions

**C** adequacy

beats the null hierarchy on held-out data: the **four impostors** (scheduler field, exogenous field, priors, convergence)

**D unfitted predictions**

reproduces statistics left out of the fit  
and the model's signature (a jump,  
 $s^{-3/2}$  sizes, a sum rule)

**E** **interventional**

fitted on one side of a natural experiment, predicts the sign and size on the other side, or across a reversal

**E** **identifiability**

synthetic recovery with the village's  
real sampling; stable over bins, embed-  
ding model, split halves

**G** ground truth

agrees with known structure: assigned  
roles and leaders, rooms, families,  
scheduler changes

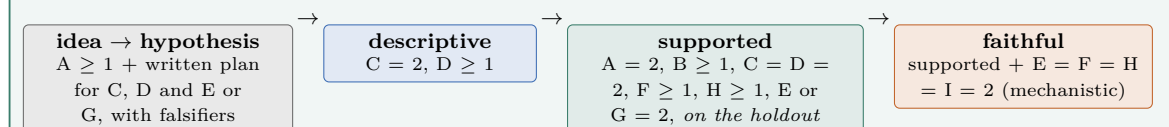
**H** comparative

beats the rival models named on the card (held-out likelihood, description length, partition contrasts)

**I** transfer

holds in other periods of the same mode and regime, **including the locked holdout**

**Promotion levels** (each step needs the one before)



Example scorecard: H111



scores 1 1 1 2 1 1 1 1 0: an unfitted prediction passed ( $D = 2$ ), but C needs a held-out null win and I needs the holdout. Level: **hypothesis**. Credence 0.65 says the claim is likely to hold; it does not say the model is the mechanism.

(c) Where the scored claims sit

